

Jyot Buch

jyotb2@illinois.edu | +1 217 819 9416 | github.com/JyotBuch

Education

- University of Illinois, Urbana-Champaign**, MSc in Statistics Aug 2024 – May 2026
- **Coursework:** Causal Inference, Deep Learning for Computer Vision, Unsupervised Learning
 - TA for BADM 554: **Adoption of GitHub Copilot and Claude Code** for data mining for business major students; **contributed** an upcoming-assignment feature to an **open-source Canvas MCP server** built by faculty.
 - Co-Founder, StatCom: **Led analytics projects for community partners**, coordinating **50+ students** across seven engagements and **presenting findings to stakeholders** to inform data-driven decisions.
- Vishwakarma University, Pune**, BTech in Computer Engineering Aug 2019 – May 2023
- **Coursework:** Data Structures and Algorithms, Database Management Systems, Operating Systems, Machine Learning

Work Experience

- Data Scientist Intern @ Amazon** May 2025 – Aug 2025
- Enhanced a retrieval service using **Python and FastAPI on AWS Lambda**, combining **BM25, dense embeddings, and metadata filters** to serve real-time queries at **p99 latency under 550ms**, improving Recall@30 by 8 points.
 - Deployed an **automated LLM prompt experimentation and optimization framework** for key-value extraction from financial and tax documents, achieving 92% field-level accuracy.
 - **Designed POC** for a query decomposition framework scoring **per-intent retrieval coverage to monitor model behavior** and **guide prompt engineering** for summarization tasks.
- Automation Engineer @ Accelirate Inc.** Dec 2023 – Jul 2024
- Extended and maintained a queue-triggered OCR and NLP invoice extraction pipeline in **UiPath Agents** and C#, processing invoices concurrently in production with PostgreSQL for structured output persistence.
 - Built a **confidence-gated routing layer and validated extraction quality** through systematic testing, sustaining 88 % field-level accuracy on 500+ invoices/day across 8 document formats.
- Data Scientist Intern @ Eaton** Aug 2022 – Dec 2022
- Designed and implemented a Drawing-to-Data extraction pipeline in Python using a **YOLOv5 model trained from scratch** on AWS, parsing BOM tables, title blocks, and revision notes **across 250K+ production drawings**.
 - Wrote tests across all pipeline stages and built a confidence-gated exception-routing layer, reducing manual review by 70% while **maintaining mAP@0.5 of 0.91 in production**.

Research Experience

- Research Scientist @ scienceNLP Lab** Jan 2026 – Present
- **Fine-tuned SapBERT** via contrastive training on newer ontologies and built an entity linking module over a 6M+ entity **FAISS index**, achieving 0.675 Recall@1 and 0.736 MRR; manuscript in preparation for *Bioinformatics*.
 - Designed a **biomedical knowledge graph** ingesting PubMed literature via REST APIs, with entity harmonization and bulk **Neo4j** loading, unifying data across 18+ relationship types; manuscript in preparation for *JAMIA*.
 - Designed an **agentic GraphRAG system over a biomedical knowledge graph** to surface latent relationships through multi-step retrieval and reasoning, supporting **open discovery and hypothesis generation** across biomedical entities.

Projects

- Revisit - Capture now. Come back informed**
- Built and shipped a full-stack research intelligence product, a **browser extension and Next.js web app**, using **LLM tool orchestration, PostgreSQL, and pgvector** to transform captured content into structured research summaries
 - Integrated LLM observability using **Langfuse tracing** and an LLM-as-judge **eval harness to measure implicit relevance feedback**, orchestrated as a daily batch job with idempotent re-run semantics.
- Interview.AI - Agent-Based Interview Assistant - HackIllinois 2025**
- Architected a **supervisor-worker multi-agent system** with Docker, designed via **LangGraph orchestration** along with **web-grounded retrieval** for real-time question generation with structured response-quality grading.
 - Delivered a full-stack interactive application with a **React frontend and Node.js backend**, enabling users to upload resumes, run mock interviews, and receive structured feedback reports

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, SQL, R, C++ , Java
- **Frameworks & Tools:** React, Next.js, Node.js, FastAPI, LangGraph, PyTorch, TensorFlow, Docker, AWS Lambda
- **AI & Data Systems:** RAG, Hybrid Retrieval, Reranking, A/B Testing, PostgreSQL, FAISS, Pinecone, pgvector, Neo4j